Practice 2. Computational Chemistry with eChem Atomic charges, bond energies, conformation, vibrations, molecular orbitals and UV/vis spectra

Chemistry. Degree in Automotive Engineering

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Objective

This practice is used to use computational chemistry tools in a controlled manner and connect the results with the course syllabus. You will work with eChem/VeloxChem notebooks already adapted for local execution. First, the whole class will use three common reference molecules: hydrogen, methane and benzene. Then, each group will study a single molecule of medium complexity, chosen because it is chemically interesting and because it has at least one dihedral that allows a rotational potential energy curve to be made.

The conceptual starting point is the Schrödinger equation. In the notes we have seen that, even in a model system as simple as a particle confined in a box, the solution of the equation does not give any possible energy, but discrete levels and wave functions with nodes. The calculations you will do here are much more complex, because many nuclei and many electrons are involved, but the logic is the same: a quantum model of the molecule is constructed, an approximate energy and electronic distribution are calculated, and from this, charges, bonds, vibrations, orbitals and spectra are interpreted.

Before opening the eChem notebooks, first run the LibreTexts interactive notebook Particle in an Infinite Potential Box. It is not part of the handout, but it is a useful check to see how a simple differential equation translates into quantized energies and wave functions. Try changing the size of the box, the mass of the particle, and the quantum number: this gives intuition to understand why, in a real molecule, changing the geometry or mass of the atoms also changes energies, vibrations, and spectra.

At the end of the practice you should be able to:

- install and verify a local Jupyter environment for eChem;
- calculate and interpret atomic charges, bond dissociation energies, conformational profiles onals, normal vibration modes, molecular orbitals and UV/vis spectra;
- perform manual calculations with the expressions worked on in the chapters of the course, especially the chapter on light, electronic structure and bonding;
- discuss which conclusions are robust and which depend on the computational model used.

Important. Calculations are a tool for chemical thinking, not a black box. In each answer you must indicate which notebook you used, which molecule you calculated, which parameters you modified and what chemical interpretation you make of the result.

Material available on the web

The notebooks will be published in two formats: an HTML version to consult them without executing anything, and an ipynb version that you can download and run with Jupyter.

As a conceptual warm-up, you will also use the external notebook Particle in an Infinite Potential Boxing, which shows the solution to the particle in an infinite box problem. This notebook does not require the eChem environment and is used to check, with a model system, what it means to solve the Schrödinger equation.

Notebook	Links	Main use
AtomicCharges	HTML / ipynb	ESP, RESP and CHELPG charges; polarity, electronegativity, resonance and functional groups.
BDEtohydrogen	HTML / ipynb	Bond dissociation energies, comparison with manual estimates of binding energies.
DihedralScan	HTML / ipynb	Potential energy curves for rotation at around a single bond; conformations and rotation barriers.
Vibrational Modes	HTML / ipynb	Geometric optimization, Hessian, vibrational frequencies and animation dynamics of normal modes.
Orbital Visualizer	HTML / ipynb	Qualitative visualization of HOMO, LUMO, ÿ, ÿ orbitals , antibonding and free pairs.
UV_Vis_spectrum	HTML / ipynb	Electronic transitions, energy of photons, wave number and effect of conjugation

There are also the environment files:

- environment-echem-macos-linux.yml
- environment-echem-windows.yml

Installation

1. Install Conda

Use Miniconda or Anaconda. If you already have Conda installed, you don't need to reinstall it. On Windows It is best to work from the Anaconda Powershell Prompt; on macOS and Linux, from the terminal.

Avoid folders with very long paths or problematic characters if you are working on Windows. For For example, a folder like C:\AutomotiveQuimica\P2 is better than a very deep path inside OneDrive.

2. Create the environment

Download the environment file corresponding to your operating system and locate it in the folder where you saved it. saved.

macOS or Linux:

```
conda env create -f environment-echem-macos-linux.yml conda activate
quimica-echem
```

Windows:

```
conda env create -f environment-echem-windows.yml conda
activate chemistry-echem
```

If the environment already existed and you just want to update it:

```
conda env update -f environment-echem-macos-linux.yml --prune
```

3. Verify the installation

Run a minimal check:

```
python -c "import veloxchem, xtb, dftd4, rdkit, py3Dmol; print('eChem OK')"
```

If eChem OK appears, you can open Jupyter:

```
jupyter-lab
```

When the browser opens, choose the quimica-echem kernel. If Jupyter was already open before activating the environment, close it and restart it from the terminal with the environment active.

Common mistakes

- conda-content-trust: If the message appears during conda activate but the prompt changes to (chem-echem), the environment has been activated anyway. This can be fixed later by updating Conda, but it does not block the practice.
- Line magic function %%capture not found: You are running a Colab cell as if it were a Python line. Use local adapted notebooks or run the entire cell as a Jupyter cell.
- xtb-python is not available or dftd4-python is not available: Install the packages within the active environment:

  ```
  conda install -c conda-forge xtb-python dftd4-python
  ```

  Then restart the kernel.
- name 'XtbCalculator' is not defined: the kernel was started before installing xtb-python. Restart Jupyter completely from the terminal with quimica-echem active.

Expressions you must use manually

Notebooks give numerical results, but the report must include manual calculations. Use units and conversion factors explicitly.

$$E = h\text{ÿ} = \frac{hc}{\text{ÿ}} \quad (1)$$

$$\tilde{\text{ÿ}} = \frac{1}{\text{ÿ}} = \frac{n}{c} \quad (2)$$

$$E_{mol} = N_A\ \text{ÿ}\ \frac{hc}{\;} \quad (3)$$

$$\text{ÿ}H_{rxn}\ \text{ÿ}\ \text{ÿ}D(\text{bonds broken})\ \text{ÿ}\ \text{ÿ}D(\text{bonds formed}) \quad (4)$$

$$OE = \frac{N_{linking}\ \text{ÿ}\ N_{antilinking}\ 2}{\;} \quad (5)$$

$$\tilde{\text{ÿ}}_{vib}\ \text{ÿ}\ \text{ÿ}\ k\ \mu \quad (6)$$

Remember that the **wavenumber** is ÿ˜ and is usually expressed in cmÿ1 . It is not a frequency: it is the inverse of the wavelength.

Working molecules

First the whole class will work on the same three small or fast molecules. Not all of them are suitable for all notebooks: that is precisely why they are good controls. Hydrogen allows us to understand the simplest covalent bond, methane gives a clean case for C–H bond energies, and benzene introduces aromaticity, delocalized *ÿ* orbitals, and UV absorption.

Common Molecule	Formula	Main use	Recommended laptops
Hydrogen	H_2	Minimum covalent bond and H–H energy	BDEtohydrogen, OrbitalVisualizer
Methane	CH_4	C–H bonds and combustion energy estimates	BDEtohydrogen, OrbitalVisualizer
Benzene	C_6H_6	Aromaticity, ÿ orbitals, HOMO–LUMO and UV	AtomicCharges, Orbital Visualizer, UV_Vis_spectrum

Then each group will work on a single molecule of medium complexity. All have a proposed dihedral so that the DihedralScan notebook can be applied. If when running the notebook the atom indices do not match the indicated dihedral, it is necessary to display the molecule with indices and choose four equivalent atoms.

Group Average molecule	Formula	Course context	Recommended Dihedral
In Phenol	C_6H_6O	Functional groups, pigments, acidity, aromaticity	C–C–O–H of the phenol group
B Toluene	C_7H_8	Benzene/toluene, vapor pressure, aromatic hydrocarbon	H–C–C–C around C(sp3)–C(aryl)
C Ethylbenzene	C_8H_{10}	Aromatic hydrocarbons, fuels, alkyl substituents	C–C–C–C between chain ethyl and ring
D Acetophenone	C_8H_8O	Aromatic carbonyl, polarity, UV	C–C(=O)–C(aryl)–C

Group	Average molecule	Formula	Course context	Recommended Diedric
E	Benzoic acid	$C_7H_6O_2$	Acid-base, carboxyl, aromaticity	O=C–O–H or C(aryl)–C(=O)–O
F	Ethyl acetate	$C_4H_8O_2$	Esters, azeotropes, strengths intermolecular	C–C–O–C or C–O–C=O
G	Cyclohexane	C_6H_{12}	Acetone-cyclohexane mixtures, conformation	C–C–C–C inside the ring
H	Methylcyclohexane	C_7H_{14}	Hydrocarbons, conformation axial/equatorial	C–C–C–C around the methyl
I	Nitromethane	CH_3NO_2	Nitro group, fuels, resonance	H–C–N–O
J	Adipic acid	$C_6H_{10}O_4$	Polyamides, materials, group carboxyl	C–C–C–C of the chain
K	1,6-hexanediamine	$C_6H_{16}N_2$	Polyamides, amines, molecular flexibility	C–C–C–C of the chain
L	Isophthalic acid	$C_8H_6O_4$	Polymers, diacids, aromaticity	C(aryl)–C(=O)–O–H of a carboxyl
M	Dimethyl isophthalate	$C_{10}H_{10}O_4$	Aromatic esters, polyester precursors	C(aryl)–C(=O)–O–C of a ester

Before starting any calculations with the molecule in the group, look it up in PubChem. Write down the name preferably, the CID, the molecular formula, the molar mass and the canonical or isomeric SMILES. Use this information to build the molecule in the notebooks and to verify that you are working with the correct structure. If PubChem returns several similar records, justify which one you chose.

Questions

Answer the following questions for the average molecule assigned to your group, **if possible.** When a question is not applicable, too slow or not very informative for the group molecule, explain it chemically and use one of the three common molecules as a reference. It has not been of forcing a tool when the result does not make chemical sense.

1. Structure and context

1. If possible for the molecule in the group, first search for it in PubChem and indicate its name preferably, the CID, the molecular formula, the molar mass and the SMILES used in notebooks.
2. If possible for the molecule in the group, write the main Lewis structure and, if necessary, the forms most relevant resonances.
3. If possible for the molecule in the group, identify the types of bonds present: ÿ, ÿ, bonds polar, lone pairs and functional groups.
4. If possible for the group's molecule, explain in which course topic the molecule appeared. or why it is useful in the context of automotive, materials, combustion, solutions, batteries or light.

2. Atomic charges

Use AtomicCharges with the group molecule, if possible.

1. If possible for the molecule in the group, compare the ESP, RESP, and CHELPG charges of the most relevant atoms.

2. If possible, for the molecule in the group, indicate which atoms are richer and poorer in electron density. Relate this to electronegativity, resonance and geometry.
3. If possible for the molecule of the group, and especially if it contains a carbonyl, carboxyl, nitro, alcohol, amine or phenol group, explain how the calculated charges justify the polarity and possible reactivity.

3. Bonding energies

Use BDEtohydrogen when there is a clear X–H bond, if possible for the molecule in the group. If your molecule does not have a suitable H or the calculation is slow, do it with hydrogen, methane, or a close reference molecule.

1. If possible for the molecule in the group, calculate a bond dissociation energy and indicate What link have you broken?
2. If possible for the group molecule, manually estimate a reaction energy with energies link media:

   ÿHrxn ÿ ÿD(broken) ÿ ÿD(formed)

3. If possible for the group molecule, compare the manual value with the computational result. Discuss why an average bond energy cannot exactly reproduce a particular molecule.

4. Conformation

Use DihedralScan for the group molecule. All assigned average molecules have a proposed dihedral; use it if possible, or choose an equivalent one after displaying the atomic indices.

1. If possible for the molecule of the group, choose a dihedral and draw or indicate the four atoms that define it.
2. If possible for the molecule in the group, represent the relative energy as a function of the angle. Ident-tify minimums and maximums.
3. If possible for the molecule in the group, relate the rotational barrier to ÿ bonds, steric hindrance, lone pair interactions, or conjugation.
4. If possible for the molecule of the group, and especially if you are working with ethyl acetate, cyclohexane or derivatives, compare your curve with the idea of alternating/eclipsed or axial/equatorial conformations.

5. Molecular vibrations

Use VibrationalModes with the optimized group molecule, if possible. If it is too slow, perform the calculation with methane or benzene and compare qualitatively with the group molecule.

1. If possible for the group molecule, run geometry optimization and vibrational analysis. Indicate whether imaginary or negative frequencies appear and what this would mean.
2. If possible for the molecule in the group, choose two normal modes and describe them: stretching of connection, angular bending, torsion or collective movement.
3. If possible for the group molecule, animate the selected modes and include captures representing sentatives or a precise description of the movement.

4. If possible for the molecule in the group, qualitatively compare the frequencies with bond strength and reduced mass. For example, discuss why an O–H or C–H stretch tends to appear at high frequencies.

6. Molecular orbitals

Use OrbitalVisualizer.

1. If possible for the group molecule, visualize HOMO and LUMO. Classify them qualitatively vaming as ÿ, ÿ, *ÿ* ÿ , free pair or combination.
2. If possible for the group molecule, compare the result with your Lewis structure and with the molecular orbital model from the light and bonding chapter.
3. If possible for the group molecule, discuss the bond order when it is informative. If not, that is, use hydrogen or benzene as a comparison.
4. If possible for the molecule of the group, and especially if it is aromatic, explain how it is distributed empty the *ÿ* orbitals on the ring.

7. UV/vis and manual calculation with photons

Use UV_Vis_spectrum when the calculation is reasonable, if possible for the molecule in the group. For molecules without informative transitions in the visible, do the calculation anyway and interpret why the absorption is in the UV.

1. If possible for the molecule in the group, note a wavelength of a relevant transition.
2. If possible for the molecule in the group, manually calculate the energy of a photon, the energy per mole and the wave number:

$$E = \frac{hc}{\ddot{y}}, \qquad \text{Emol} = \text{NA}\ \ddot{y}\ \frac{hc}{}, \qquad \ddot{y}^{\sim} = \frac{1}{\ddot{y}}$$

3. If possible for the group molecule, compare saturated, carbonyl and aromatic molecules. Relate the position of the absorption to the HOMO–LUMO separation and conjugation.
4. If possible for the molecule in the group, and especially if it is related to pigments or color, explain why the small molecules in the course do not normally absorb visible light.

Delivery

Each group will deliver a short report in PDF. Before writing it, consult the guide to writing internship reports: It explains how to structure the report, how to present figures and tables, and how to correctly cite the sources used. The report for this internship must include:

- names of group members and group letters;
- table of molecules worked on, including the three common ones and the group molecule, notebooks executed and parameters modified;
- numbered answers to the previous questions;
- exported captures or figures of the most important results;
- manual calculations with units;

- a final discussion of the limitations of the calculations.

It is not necessary to include all the outputs from the notebook. Select the data that you use to justify a chemical response.

References

- eChem book: Fransson, T. et al. *Computational Chemistry from Laptop to HPC: A Note-book Exploration of Quantum Chemistry.* KTH Royal Institute of Technology, 2022. DOI: 10.30746/978-91-988114-0-7.

- eChem website: https://kthpanor.github.io/echem/
- eChem installation: https://kthpanor.github.io/echem/docs/install.html
- eChem colabs: https://kthpanor.github.io/echem/colab/
- Course materials: chapters on gases, intermolecular forces, combustion, batteries, light and materials.